

MATPLOTLIB TASKS

1. Load the Excel dataset using Pandas and display first 10 rows.
2. Display dataset shape and create a simple bar chart showing total rows vs total columns.
3. Display column names and plot a bar chart showing length of each column name.
4. Check data types of all columns and visualize numeric vs non-numeric column count.
5. Convert `signup_date` column to datetime format and plot number of signups per year.
6. Check for missing values in each column and visualize missing values using bar chart.
7. Count total number of customers and create a single bar chart representing total count.
8. Count number of unique cities and visualize top 5 cities using bar chart.
9. Count number of unique states and plot state distribution using pie chart.
10. Display only `first_name`, `email`, `city` and visualize city frequency using horizontal bar chart.
11. Filter customers whose age is greater than 25 and plot age distribution histogram.
12. Filter customers from a specific state and plot their age distribution.
13. Filter customers who signed up after a specific year and plot signup trend.
14. Sort customers by age (ascending) and plot line graph of sorted ages.
15. Sort customers by `signup_date` (descending) and plot signup timeline.
16. Filter customers whose email contains "gmail" and compare count vs non-gmail using bar chart.
17. Filter customers whose first_name starts with 'A' and plot their city distribution.
18. Filter customers between age 20 and 35 and compare with other age groups using bar chart.
19. Display top 10 youngest customers and plot their ages using line graph.
20. Display top 10 oldest customers and visualize their age comparison using bar chart.
21. Calculate mean, median, std of age using NumPy and visualize age distribution using histogram
22. Find minimum and maximum age using NumPy and highlight them on a bar chart
23. Calculate 25th, 50th, 75th percentile of age and visualize using boxplot
24. Normalize age column using NumPy formula and plot normalized age distribution
25. Create age frequency distribution using NumPy and plot bar chart
26. Calculate variance of age and compare actual vs mean age using line graph
27. Identify customers above average age and visualize comparison bar chart
28. Create age bins using NumPy and visualize grouped histogram
29. Calculate z-score of age and plot scatter plot of z-scores
30. Compare age distribution between two states using histogram
31. Extract signup year and calculate yearly customer growth using Pandas, compute growth % using NumPy, plot line graph
32. Extract signup month and visualize monthly trend using bar chart
33. Calculate cumulative customer count per year using NumPy and plot cumulative line graph
34. Compare signup trend between top 3 states using multi-line graph
35. Calculate moving average of monthly signups using NumPy and plot trend
36. Identify peak signup month and highlight it in graph
37. Compare weekday vs weekend signups and visualize using bar chart
38. Create pivot table (year vs state), convert to NumPy array and visualize heatmap
39. Calculate yearly growth rate and visualize using line graph

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40. Create subplot dashboard (Yearly trend + Monthly trend)
41. Calculate customer count per city and plot horizontal bar chart
42. Find top 5 cities and visualize percentage share using pie chart
43. Calculate average age per state and plot bar chart
44. Compare average age of top 3 states using grouped bar chart
45. Create stacked bar chart (state vs age group)
46. Calculate state-wise age variance using NumPy and visualize
47. Identify city with maximum customers and highlight in bar graph
48. Compare top 3 cities age distribution using multiple histograms
49. Create scatter plot of city index vs average age
50. Create 2x2 subplot dashboard (City Count, State Count, Avg Age, Age Histogram)
51. Create correlation matrix using Pandas and visualize using imshow
52. Calculate age density distribution using NumPy and plot
53. Identify outliers in age using IQR method and visualize
54. Compare mean vs median age per state using grouped bar chart
55. Create cumulative age distribution graph
56. Calculate standard deviation per state and visualize
57. Create boxplot comparing age distribution across states
58. Calculate percentage contribution of each state and visualize donut chart
59. Create ranking based on signup count and visualize ranking chart
60. Compare customers above 30 vs below 30 using pie chart
61. Create complete demographic dashboard (Age histogram + State bar + Signup trend)
62. Create advanced pivot heatmap (State vs Age Group)
63. Calculate growth trend and overlay moving average
64. Compare top 5 cities yearly signup trend
65. Create regression-style scatter plot (Signup year vs count)
66. Build multi-line graph comparing state-wise growth
67. Create final analytics dashboard with 4 subplots combining:
 - Age distribution
 - State distribution
 - Signup yearly trend
 - Top cities comparison

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PROJECTS

➡ Customer Demographic Analytics Dashboard

Description : Build a demographic analysis dashboard to understand customer age distribution and geographic spread.

- ❖ Load and clean the dataset
- ❖ Calculate mean, median, and standard deviation of age using NumPy
- ❖ Create age groups (18–25, 26–35, 36+)
- ❖ Count customers per state and city using Pandas
- ❖ Identify top 5 cities
- ❖ Build a 2x2 subplot dashboard:
 - Age histogram
 - State distribution bar chart
 - Top cities horizontal bar chart
 - Age group pie chart
- ❖ Add titles, legends, grid, and styling

➡ Customer Signup Growth Analysis

Description : Build a demographic analysis dashboard to understand customer age distribution and geographic spread.

- ❖ Convert signup_date to datetime
- ❖ Extract year and month
- ❖ Calculate yearly signup count
- ❖ Compute growth percentage using NumPy
- ❖ Calculate moving average
- ❖ Plot:
 - Yearly trend line
 - Growth percentage line
 - Moving average overlay
 - Highlight peak growth year
- ❖ Highlight peak growth year.

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➡ Age Segmentation Study

Description : Segment customers based on age groups and analyze behavior across regions.

- ❖ Create age bins using NumPy
- ❖ Generate age group column
- ❖ Create pivot table (State vs Age Group)
- ❖ Visualize using stacked bar chart
- ❖ Detect outliers using IQR method
- ❖ Plot boxplot for age groups
- ❖ Compare young vs older customer distribution

➡ City Dominance & Contribution Analysis

Description : Identify dominant cities and measure their contribution to overall customer base.

- ❖ Rank cities by customer count
- ❖ Calculate percentage contribution

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- ❖ Identify top 10 cities
- ❖ Plot horizontal bar chart
- ❖ Plot pie chart for top cities contribution
- ❖ Compare yearly growth for top cities
- ❖ Create multi-line city comparison graph.

➡ Complete Customer Intelligence Dashboard

Description :Build a professional executive-level dashboard showing key demographic and growth metrics.

- ❖ Perform data cleaning
- ❖ Feature engineering (age_group, signup_year, signup_month)
- ❖ Create correlation matrix
- ❖ Build 4-chart dashboard:
 - Age distribution
 - State distribution
 - Yearly growth trend
 - Top cities comparison
- ❖ Add annotations and professional styling.

➡ Customer Growth Forecasting System

Description :Build a basic forecasting system to predict future customer growth trends.

- ❖ Extract yearly signup trend
- ❖ Calculate yearly growth rate using NumPy
- ❖ Compute rolling average
- ❖ Manually compute regression-style trend line
- ❖ Forecast next year signup (basic projection)
- ❖ Plot actual vs predicted trend
- ❖ Highlight predicted value

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➡ Advanced Customer Segmentation Model

Description :Create an analytical segmentation system based on normalized age metrics.

- ❖ Normalize age using NumPy
- ❖ Create segmentation score
- ❖ Divide customers into segments (Low, Medium, High)
- ❖ Compare segment distribution across states
- ❖ Create grouped bar charts
- ❖ Build multi-subplot comparison dashboard

➡ Seasonal Signup Intelligence System

Description :Analyze seasonal and weekday signup behavior patterns.

- ❖ Extract month and weekday from signup_date
- ❖ Calculate monthly signup distribution
- ❖ Compare weekday vs weekend signups
- ❖ Create heatmap (Month vs State)
- ❖ Identify peak seasonal patterns
- ❖ Add annotated visualization

➡ Strategic Demographic Decision Model

Description :Build a decision-support model to identify high-growth and aging regions.

- ❖ Identify fastest growing state
- ❖ Find state with highest average age
- ❖ Compare youth-dominant vs senior-dominant states
- ❖ Create growth index formula using NumPy
- ❖ Rank states based on strategic index
- ❖ Build executive-level 4-chart summary dashboard.